7003

C20-A-AA-AEI-CH-CHST-BM-TT-MET-MNG-C-CM-EC-EE-M-CHOT-CHPC-CHPP-PET-AMT-AMG-WD-CAI-AIM-CCB-CCN–103

7003

BOARD DIPLOMA EXAMINATION, (C-20)

MARCH/APRIL—2025

FIRST YEAR (COMMON) EXAMINATION

ENGINEERING PHYSICS

Time : 3 Hours] [*Total Marks* : 80

PART—A

3×10=30

Instructions : (1) Answer **all** questions.

(2) Each question carries **three** marks.

(3) Answers should be brief and straight to the point and shall not exceed five simple sentences.

1. Write fundamental quantities in SI system with their units.

2. State the polygon law of vectors.

3. A ball is thrown into the air with a velocity of 150 m/s at an angle of 30° to the horizontal. Find the time of flight (g = 10 m/s^2).

4. Define friction and write any two advantages of it.

5. Derive the relation between kinetic energy and momentum.

6. State the laws of the simple pendulum.

7. State the Charle's 1st and 2nd laws of gas.

8. Define reverberation time. Write Sabine's formula.

9. Define specific resistance. Write its SI unit.

10. Define magnetic field and magnetic induction field strength.

PART—B

8×5=40

Instructions : (1) Answer **all** questions.

(2) Each question carries **eight** marks.

(3) Answers should be comprehensive and criterion for valuation is the content but not the length of the answer.

11. *(a)* Define cross product. Write any six properties of cross product. 2+6

(OR)

(b) Define the horizontal projection. Show that the path of the horizontal projectile is a parabola. 1+7

12. *(a)* Derive an expression for the acceleration of a body moving up on a rough inclined plane. 8

(OR)

(b) A machine gun fires 300 bullets per minute. Each bullet moves with a velocity of 500 m/s. If the mass of each bullet is 2 gram, find the power of the gun. 8

13. *(a)* Derive an expression for acceleration and time period of a particle executing simple harmonic motion. 5+3

(OR)

(b) Define molar specific heat of a gas and derive $C_p - C_v = R$. 2+6

14. *(a)* State the conditions of good auditorium. A boy hears an echo of his own voice from a distant hill after 2 seconds. If the velocity of sound is 340 m/s, find the distance of the hill. 4+4

(OR)

(b) Define viscosity and give three examples. Write the Newton's formula for viscous force. 2+3+3

15. *(a)* Derive an expression for magnetic induction field strength at a point on the equatorial line of a bar magnet placed in a uniform magnetic field. 8

(OR)

(b) State and explain photoelectric effect. Write three applications of photoelectric cell. 5+3

PART—C

10×1=10

Instructions : (1) Answer the following question.

(2) The question carries **ten** marks.

(3) Answer should be comprehensive and the criterion for valuation is the content but not the length of the answer.

16. Derive an expression for maximum height, time of ascent and horizontal range in oblique projection. 3+3+4

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